

Diabetes Risk Prediction

A supervised machine-learning study on 70,693 BRFSS patient records

GROUP

Ali Raza · 2540010
Muhammad Ammar · 2540008
Taha Ali · 2540008

INSTRUCTOR

Sir Zaki

SEMESTER

Semester 2 · 2026

The Problem and Our Dataset

Why this project

- Diabetes affects 1 in 11 adults worldwide
- Early detection saves lives and lowers cost
- Survey data can flag high-risk patients early
- Our model learns these patterns from real data

DATASET AT A GLANCE

Source	CDC BRFSS 2015 (Kaggle)
Records	70,693 patients
Features	21 health indicators
Target	Diabetes_binary (0 / 1)
Balance	Balanced 50 / 50
Task	Binary Classification

Objectives & Measurable Goals

Objectives · what we will do

- Load the dataset and describe its structure
- Clean data — duplicates, dtypes, missing values
- Run EDA and analyse correlation with the target
- Train a Random Forest classifier (supervised)
- Evaluate accuracy, confusion matrix, importance

Goals · how we measure success

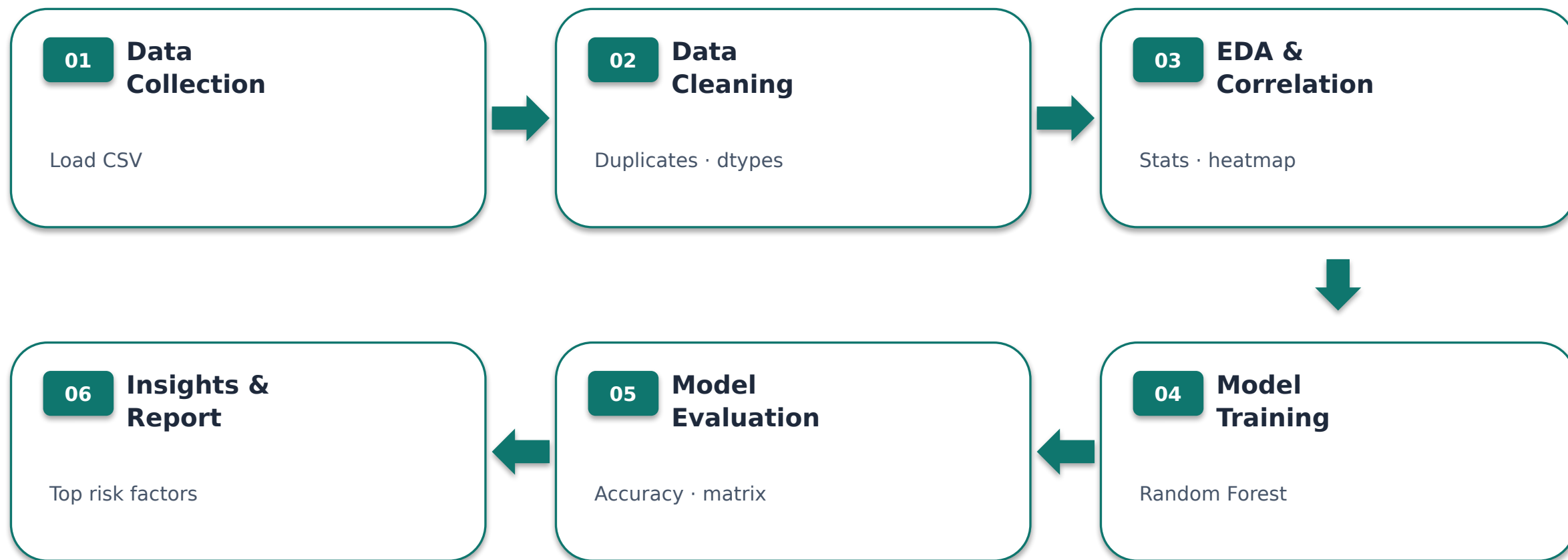
- Reach $\geq 80\%$ test accuracy
- Surface top 5 diabetes risk factors
- Deliver notebook + final presentation

LEARNING TYPE

Supervised Learning

Target is labeled (0 / 1) → Binary Classification

Project Block Diagram



Tools, Timeline & Deliverables

Tools & Technologies

- **Python 3** Core language
- **pandas · NumPy** Data handling
- **matplotlib · seaborn** Visualization
- **scikit-learn** Random Forest model
- **Jupyter Notebook** Development & report

FOUR-WEEK TIMELINE

WEEK 1

Load · clean · describe

WEEK 2

EDA · correlation · plots

WEEK 3

Train Random Forest · evaluate

WEEK 4

Report · slides · viva prep